

OPIM 201 – Practice Problem Set 4

1. An integrated petroleum company has a single unloading facility at its Singapore refinery. Due to random variations in weather, loading delays, and other factors, ships arriving at the refinery to unload crude oil arrive according to a probability distribution with an average arrival rate of 8 ships per week and CV_a equal 1. The unloading facility has an average service rate of 12 ships per week with CV_p equal 1.
 - (a) What is the average number of ships waiting to gain access to the single unloading facility?
 - (b) What is the average total time (waiting plus actual delivery) that a ship spends at the refinery?
 - (c) The company is considering a second unloading berth that could be rented for \$6,000 per week. The service time for this berth would also be exponential ($CV_p=1$) with the same service rate of 12 ships per week as the company's own berth. What would be the resulting average total time a ship would wait? How should the company assess this option of renting a second berth?
2. **Queuing Theory in Court (Based on an actual case)** A worker at the Unemployment Office is responsible for processing a company's forms when it opens for business. The worker can process an average of 4 forms per week. In 2011 an average of 1.8 companies per week submitted forms for processing, and the worker had an order flow time of 0.45 weeks. In 2012 an average of 3.9 companies per week submitted forms for processing and the worker had a 5 week order flow time. The poor worker was fired, but later sued to get her job back. The court said that because the amount of work submitted to the worker had approximately doubled, the worker's backlog also should have doubled. Because her backlog increased by more than a factor of 10, she must have been slacking off, so the state was justified in firing her. Assume $CV_a=1$ and $CV_p=1$. Use the concepts and tools learned in OPIM 201 to defend the worker.
3. Pharma Dev develops and markets new technological products to be used in health care. The development of a new product operates as follows. When a new technology meets the requisite market potential, a new patent is filed. Patents are granted for a period of 12 years starting from the date of issue. Once the patent is filed, the new technology is developed at one of its three independent development centers, and is then launched to the market. Each product is developed at only one center, and each center can only develop a single patented technology at a time. On average, Pharma Dev files a new patent every 5 months (with standard deviation of 5 months). The average development process lasts 12 months (with standard deviation of 12 months).
 - (a) What is the utilization of Pharma Dev's development facilities?
 - (b) Development Time: How long does it take for an average technology from winning a patent to start the development process?
 - (c) Market Life: How many years of patent life are left for an average product launched to the market?
 - (d) On average how many patented products are undergoing development or waiting to be developed?
 - (e) Pharma Dev is considering leasing an additional development center to shorten the time to market of patented products. If this center is used in addition to the three centers, how will the total time-to-market (waiting time plus development time) of a patented product change?

4. An online retailer serves customers from four regions – regions A, B, C & D. The customers from each of these regions send emails to a help desk of the online retailer every 140 seconds on average; i.e., the average email request rate is $1/140$ customers per second per region. The standard deviation of the inter-arrival time in each region is also 140 seconds. (Note that $4 \cdot (1/140) = 1/35$.) The online retailer has four employees answering emails. It takes on average 2 minutes to write a response email. The standard deviation of the processing times is 1 minute.
- (a) Suppose each server is dedicated to exactly one region, i.e., each server is responsible for the email response for exactly one region. Estimate the average waiting time of each customer before being served.
 - (b) Suppose two servers are dedicated to regions A & B while the other two servers are dedicated to regions C & D. Estimate the average waiting time of each customer before being served, assuming $CV_a = 1$.
 - (c) Estimate the average waiting time of each customer before being served if all four servers serve the four regions together, assuming $CV_a = 1$.

Solution

1.

- (a)
- $a = 1/8$
- weeks,
- $p = 1/12$
- weeks,
- $m = 1$
- ,
- $CV_a = 1$
- ,
- $CV_p = 1$

$$u = 8/12 = 2/3$$

$$T_q = \frac{1}{12} \left(\frac{2^{\sqrt{4}-1}}{3} / \frac{1}{3} \right) \frac{1+1}{2} = \frac{1}{6} \text{ weeks}$$

$$I_q = \frac{1}{a} T_q = 8 \times \frac{1}{6} = 1 \frac{1}{3} \text{ ships}$$

- (b)
- $T = T_q + p = \frac{1}{6} + \frac{1}{12} = \frac{1}{4}$
- weeks

- (c)
- $u = \frac{1/12}{1/8 \times 2} = \frac{1}{3}$

$$T_q = \frac{1}{2 \times 12} \left(\frac{1^{\sqrt{6}-1}}{3} / \frac{2}{3} \right) \frac{1+1}{2} = 0.0127 \text{ weeks}$$

To assess the option of renting a second berth, the company should compare the weekly benefits resulting from the reduction in T_q against the weekly rent of \$6,000.

2. The conclusion of the court was that the worker had been slacking off, meaning her service rate had decreased with the increased workload. To defend the worker, we compute T_q with a processing rate of 4 forms/week and the new demand rate of 3.9 per week.

Backlog = average waiting time in the system (T)

Given $m=1$, $a=1/3.9$ weeks, $p=1/4$ weeks, $CV_A=1$ and $CV_P=1$, we have

$$T = T_q + p = \frac{p}{m} \frac{u^{\sqrt{2(m+1)}-1} CV_a^2 + CV_p^2}{1-u} + p = p \frac{3.9p}{1-3.9p} + p = \frac{p}{1-3.9p} = 10 \text{ weeks}$$

Hence, the expected flow time with the new (doubled) demand rate is 10 weeks, which is a lot more than the actual flow time of 5 weeks. The worker should not have been fired. She should have been given a raise!

3.

- a) There are three facilities, i.e. $m=3$. The development process takes 12 months, so $p=12$ months. On average a new patent arrives every 5 month so $a=5$ months. Therefore: $u = p/(a \times m) = 12/(5 \times 3) = 0.8$ or 80%.

- b) $p = 12$ months; $a = 5$ months; $m=3$; $u = 0.8$

$$CV_a = 5/5 = 1; CV_p = 12/12 = 1.$$

The problem is asking for the average waiting time in the line which is:

$$T_q = \frac{p}{m} \frac{u^{\sqrt{2(m+1)}-1} CV_a^2 + CV_p^2}{1-u} = \frac{12}{3} \frac{0.8^{\sqrt{2(3+1)}-1} 1^2 + 1^2}{1-0.8} = 13.3 \text{ months}$$

- c) Total Patent Life = 12 years = 144 months

$$\begin{aligned} \text{Time spent in the system (T)} &= \text{time spent in the queue (T}_q\text{)} + \text{development time (p)} \\ &= 13.3 + 12 = 25.3 \text{ months} \end{aligned}$$

$$\text{Hence patent life left} = 144 - 25.3 = 118.7 \text{ months}$$

- d) $T = 25.3$ months. By Little's law, we have $I = T/a$, hence $I = 25.3/5 = 5.06$ products.

- e) Utilization becomes $u = 12/(5 \times 4) = 0.6$; $m=4$

$$T_q = \frac{p}{m} \frac{u^{\sqrt{2(m+1)}-1} CV_a^2 + CV_p^2}{1-u} = \frac{12}{4} \frac{0.6^{\sqrt{2(4+1)}-1} 1^2 + 1^2}{1-0.6} = 2.5 \text{ months}$$

$$T = T_q + p = 2.5 + 12 = 14.5 \text{ months.}$$

So the average time to market decreases by $25.3 - 14.5 = 10.8$ months.

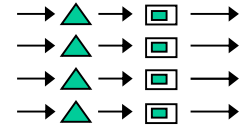
4.

a) For each of the four regions:

$$a = 140, p = 120, m = 1, CV_a = 1, CV_p = 0.5$$

$$\text{Utilization} = p/a = 0.857$$

$$T_q = \frac{120}{1} \frac{0.857 \sqrt{2(1+1)} - 1}{1 - 0.857} \frac{1^2 + 0.5^2}{2} = 450 \quad (\text{Flow Time} = 570)$$



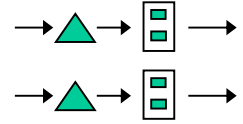
b) For each of the two submarkets :

$$a = 70, p = 120, m = 2, CV_a = 1, CV_p = 0.5$$

(Note that the inter-arrival rate is doubled when two regional markets are pooled, and thus the inter-arrival time is halved)

$$\text{Utilization} = p/(m \cdot a) = 0.857$$

$$T_q = \frac{120}{2} \frac{0.857 \sqrt{2(2+1)} - 1}{1 - 0.857} \frac{1^2 + 0.5^2}{2} = 209.94 \quad (\text{Flow Time} = 329.94)$$

c) $a = 35, p = 120, m = 4, CV_a = 1, CV_p = 0.5$

$$\text{Utilization} = p/(m \cdot a) = 0.857$$

$$T_q = \frac{120}{4} \frac{0.857 \sqrt{2(4+1)} - 1}{1 - 0.857} \frac{1^2 + 0.5^2}{2} = 94.05 \quad (\text{Flow Time} = 214.05)$$

